

Portfolio Construction

INTRODUCTION

This chapter introduces techniques to bridge the gap between portfolio construction and trading. We introduce a quantitative framework to determine the appropriate “optimal” execution strategy given the “optimal” portfolio on the efficient investment frontier.

Portfolio optimization is the process of determining an optimum mix of financial instruments. These consist of portfolios with the highest return for a specified level risk and the least risk for a specified return. These optimal portfolios are determined through advanced mathematical modeling approaches such as quadratic programming, and more recently conic optimization.

Markowitz (1952) presented a quantitative process to construct efficient portfolios through optimization. The set of all efficient portfolios constitutes what Markowitz coined the Efficient Frontier. Sharpe (1964) expanded on the efficient frontier concept by providing investors with a means to determine the most appropriate efficient portfolio on the frontier. The technique used by Sharpe was based on maximizing investor economic utility (investor happiness). Sharpe further introduced the industry to the capital asset pricing model (CAPM) which in the simplest forms is a technique to combine the market portfolio with a risk-free asset to further improve the set of risk-return above the efficient frontier. CAPM also provided the industry with metrics to quantify and manage risk, allocate investment dollars, etc. This ground breaking work by Markowitz and Sharpe paved the way for Roll and Ross (1980) with arbitrage pricing theory, Black and Litterman (1992) with alternative portfolio optimization techniques, Fama and French (1992, 1993) with their three factor model, and Michaud and Michaud (1998) with their portfolio resampling using Monte Carlo methods. Unfortunately, not as much attention has been given to portfolio construction with transaction costs. But as we show in this chapter, Markowitz and Sharpe have also paved the way to determine the most appropriate best execution strategy given the investment objectives of the fund.